

Table 1: How private, public and consortium blockchains differ from each other

Feature	Private blockchain	Public blockchain	Consortium blockchain
Access	Private	Public	Public/private
Consensus	Organisation based	Public	Selected nodes
Efficiency	High	Low	High
Centralisation	Yes	No	Partial
Consensus process	Permission based	Permission based	Permissionless
Immutability	Not completely tamper-proof	Completely tamper-proof	Not completely tamper-proof

transactions and also make it possible to do double-spending.

Types of blockchain architecture

Now that we have understood the core components of any blockchain network, let's learn about the types of blockchain architecture.

Public blockchain architecture:

In a public blockchain architecture, anyone can participate in the network. The transaction's public information is available to everyone. However, this doesn't mean that the private data of a transaction is available anymore. Examples of public blockchain architecture include Bitcoin, Lite-On and Ethereum.

Private blockchain architecture:

Not everyone can access the blockchain. The administrator or the ruling set of nodes determines who can join the network.

Federated/consortium blockchain architecture: The last type of blockchain architecture is the federated/consortium blockchain architecture. It combines the best features of both the public and private blockchains. It is also heavily controlled and is best suited for enterprise blockchains.

To understand the differences between the three, just go through Table 1.

How a blockchain works

How a blockchain works can be understood by going through a step-by-step process. The key components of a blockchain include the following:

- Node
- Block
- Transaction
- Chain
- Consensus
- Miners

Step 1: In the first step, a transaction is requested. The transaction can be either to transfer information or some asset of monetary value.

Step 2: A block is created to represent the transaction. However, the transaction has not been validated yet.

Step 3: The block with the transaction is now sent to the network nodes. If it is a public blockchain, it is sent to each node. Each block consists of the data, the previous block hash and the current block hash.

Step 4: The nodes now start validating the transaction according to the predetermined consensus method used. In the case of Bitcoin, proof-of-work (PoW) is used.

Step 5: After successful validation, the node now receives a reward based on its effort.

Step 6: The transaction is now complete.

Consensus methods

Consensus methods are integral to any blockchain type. They determine how fast, efficient and secure a transaction is.

Right now, there are various consensus methods out there. For the sake of simplicity, we are covering only the four most popular ones.

- Proof-of-work
- Proof-of-stake

- Delegated proof-of-stake
- Practical Byzantine fault tolerance

Proof-of-work (PoW): This is the first-ever consensus method that was used by a blockchain network. It was introduced with Bitcoin. In the consensus method, there are miners who are responsible for validating a transaction. The hash of a new block needs to be found so that it can be added to the network. The first one who finds it gets rewarded. This process requires extreme computational power and also has high-end requirements when it comes to hardware.

Proof-of-stake (PoS): This is used by second generation blockchain networks like Ethereum. It has a completely different approach as it doesn't involve extensive power consumption. Here coins are staked by the nodes. Ethereum has its own requirements regarding how coins need to be staked for it to be eligible to take part in the consensus algorithm. The nodes that have more coins staked have more chances of getting rewarded. This also makes proof-of-stake investment heavy.

Delegated proof-of-stake (DPoS): This is different from PoS. It handles the node selection differently. In this case, the coin holders select the nodes to take part in the consensus method. They can also vote while choosing or kicking a node. It is more suited to an established network that operates under higher levels of trust.

Practical Byzantine fault tolerance (PBFT): This is a way of solving the